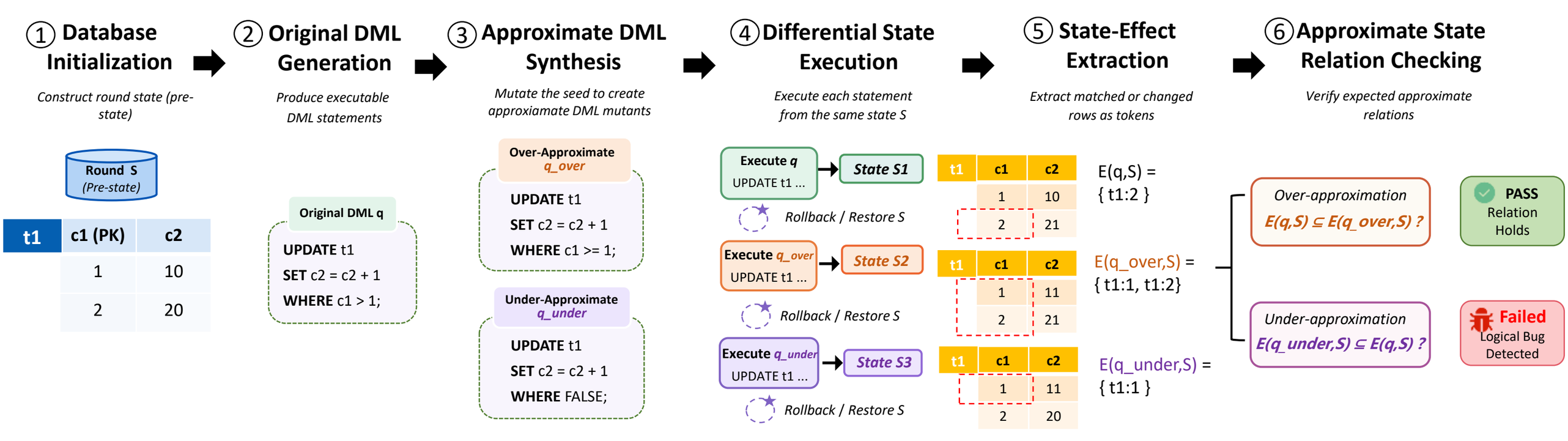
① Database Initialization
Construct round state (pre-state)
Round S (Pre-state)
t1 | c1 (PK) | c2
1 | 10
2 | 20
② Original DML Generation
Produce executable DML statements
Original DML q
UPDATE t1
SET c2 = c2 + 1
WHERE c1 > 1;
③ Approximate DML Synthesis
Mutate the seed to create approxiamate DML mutants
Over-Approximate q_over
UPDATE t1
SET c2 = c2 + 1
WHERE c1 >= 1;
Under-Approximate q_under
UPDATE t1
SET c2 = c2 + 1
WHERE FALSE;
④ Differential State Execution
Execute each statement from the same state S
Execute q
UPDATE t1 ...
State S1
Rollback / Restore S
Execute q_over
UPDATE t1 ...
State S2
Rollback / Restore S
Execute q_under
UPDATE t1 ...
State S3
Rollback / Restore S
⑤ State-Effect Extraction
Extract matched or changed rows as tokens
t1 | c1 | c2
1 | 10
2 | 21
E(q,S) = { t1:2 }
t1 | c1 | c2
1 | 11
2 | 21
E(q_over,S) = { t1:1, t1:2}
t1 | c1 | c2
1 | 11
2 | 20
E(q_under,S) = { t1:1 }
⑥ Approximate State Relation Checking
Verify expected approximate relations
Over-approximation
E(q,S) ⊆ E(q_over,S) ?
PASS
Relation Holds
Under-approximation
E(q_under,S) ⊆ E(q,S) ?
Failed
Logical Bug Detected